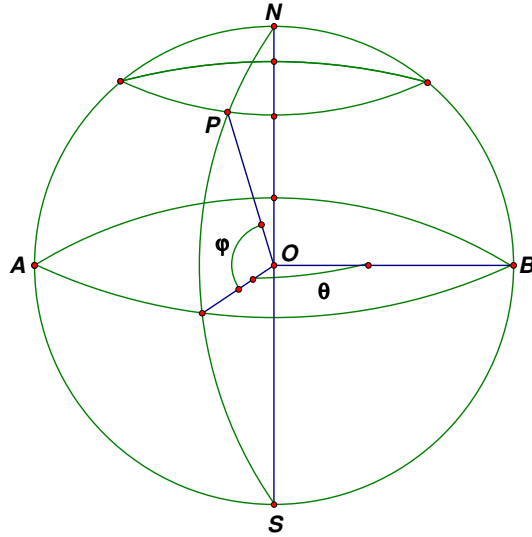


Some Spherical Geometry

A *great circle* on a sphere is the intersection of the sphere with a plane through its centre.



In the diagram we have a sphere of radius 1, AB (a great circle) is the equator, N and S are respectively the north and south pole, ϕ the *latitude* (from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$), θ the *longitude* (from $-\pi$ to π).

The position vector of a point P on a sphere of radius 1 is as follows.

$$\vec{r}(t) = (\cos \phi(t) \cos \theta(t), \cos \phi(t) \sin \theta(t), \sin \phi(t)).$$

Note both ϕ and θ are functions of t .

Taking derivative, we have

$$\begin{aligned} \vec{r}'(t) = & \left(-\sin \phi(t) \frac{d\phi}{dt} \cos \theta(t) - \cos \phi(t) \sin \theta(t) \frac{d\theta}{dt}, \right. \\ & \left. -\sin \phi(t) \frac{d\phi}{dt} \sin \theta(t) + \cos \phi(t) \cos \theta(t) \frac{d\theta}{dt}, \cos \phi(t) \frac{d\phi}{dt} \right). \end{aligned}$$

Hence the speed

$$|\vec{r}'(t)| = \sqrt{\left(\frac{d\phi}{dt} \right)^2 + \cos^2 \phi(t) \left(\frac{d\theta}{dt} \right)^2}.$$

We now want to show a great circle is the shortest distance between the points on the circle. We may assume we start from the north point with $t=0$ and conclude at t_1 ,

with $\phi(t_1) = \phi_1, \theta(t_1) = \theta_1$, and $\phi(0) = \frac{\pi}{2}$, yet $\theta(0)$ undefined (does not matter).

The distance travelled is

$$L = \int_{t=0}^{t_1} |\vec{r}'(t)| dt.$$

Note

$$|\vec{r}'(t)| \geq \sqrt{\left(\frac{d\phi}{dt}\right)^2} = \left|\frac{d\phi}{dt}\right| \geq -\frac{d\phi}{dt}$$

Hence

$$L \geq \int_{t=0}^{t_1} -\frac{d\phi}{dt} dt = \phi(0) - \phi(t_1) = \frac{\pi}{2} - \phi_1$$

The right hand side is the length of the great circle arc connecting the two points.

We now want to compute the distance between two points.

$$P = (\cos \phi_1 \cos \theta_1, \cos \phi_1 \sin \theta_1, \sin \phi_1),$$

$$Q = (\cos \phi_2 \cos \theta_2, \cos \phi_2 \sin \theta_2, \sin \phi_2).$$

Hence

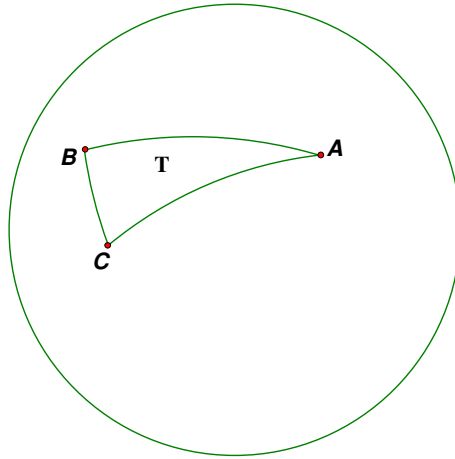
$$\begin{aligned} \overrightarrow{OP} \cdot \overrightarrow{OQ} &= \cos \phi_1 \cos \theta_1 \cos \phi_2 \cos \theta_2 + \cos \phi_1 \sin \theta_1 \cos \phi_2 \sin \theta_2 + \sin \phi_1 \sin \phi_2 \\ &= \cos \phi_1 \cos \phi_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) + \sin \phi_1 \sin \phi_2 \\ &= \cos \phi_1 \cos \phi_2 (\cos(\theta_1 - \theta_2)) + \sin \phi_1 \sin \phi_2, \end{aligned}$$

with the angle

$$\begin{aligned} &= \arccos(\overrightarrow{OP} \cdot \overrightarrow{OQ}) \\ &= \arccos(\cos \phi_1 \cos \phi_2 (\cos(\theta_1 - \theta_2)) + \sin \phi_1 \sin \phi_2). \end{aligned}$$

Distance for a sphere with radius R

$$= R \times \arccos(\cos \phi_1 \cos \phi_2 (\cos(\theta_1 - \theta_2)) + \sin \phi_1 \sin \phi_2)$$

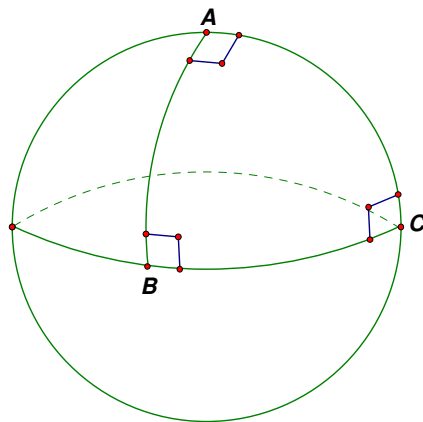


A *spherical triangle* consists of 3 arcs of 3 great circles.

Let $A^\circ, B^\circ, C^\circ$ be the angles at the vertices (or more precisely between the tangents to the arcs at each vertex). Let S be the surface area of the sphere and T the area of the triangle.

Theorem: (A Girard, 1629) $A + B + C = 180 \left(1 + \frac{4T}{S} \right)$.

Example: The triangle shown has 3 right-angles.



Meanwhile T occupies a quarter of the northern hemisphere and $T / S = 1/8$.
Therefore $180(1 + 4T / S) = 180 \times 3 / 2 = 270 = A + B + C$.

Example: If T gets smaller and smaller compared with S (like a small triangle on the

surface of the Earth) and the sum of angles tends to 180° .

The *A-lune* to be the region between the 2 great circles through A , and let α denote its area. Similarly let β, γ denote the areas of the *B-lune*, *C-lune*.

Lemma: $\frac{\alpha}{S} = \frac{A}{180}.$

Looking down on S from above A , $\frac{\alpha}{S} = \frac{2A}{360} = \frac{A}{180}.$

Proof of the Theorem: The 3 lunes cover the whole sphere (needs some imagination), but cover the triangle 3 times, which is 2 times too many, and the same with the antipodal triangle (opposite to the top triangle). Therefore $\alpha + \beta + \gamma = S + 4T$. Hence from the lemma

$$\frac{A + B + C}{180} = \frac{\alpha + \beta + \gamma}{S} = 1 + \frac{4T}{S}.$$

Finally we need some knowledge of solid angle

Solid angle $\Omega = \frac{A}{r^2}$, where A is the area on a sphere of radius r covered by a light cone from the centre.

Or using the integral definition $\Omega = \iint_S \sin \phi d\theta d\phi.$

S integral over the surface

ϕ the colatitude, the angle from the centre line to the outside region.

θ the longitude

Hence for a conical surface,

$$\begin{aligned} \Omega &= \int_0^\zeta \int_0^{2\pi} \sin \phi d\phi d\theta \\ &= \int_0^{2\pi} d\theta \int_0^\zeta \sin \phi d\phi \\ &= 2\pi(1 - \cos \zeta). \end{aligned}$$

References

1. Timothy G. Feeman, *Portrait of the Earth A Mathematician Looks at Map*, AMS, USA, 2002.
2. Christopher Zeeman, *Three Dimensional Theorems For Schools*, The Mathematical Association, UK, 2005.
3. Ben Kravitz, Lecture on Solid Angle, Preprint
4. Various Olympiad Problem Books

Exercises

1. (Warm-Up) Assume Hong Kong's position is $22^{\circ}N, 114^{\circ}E$, with the radius of the Earth 6371 km. (a) The position of Havana is $22^{\circ}N, -82^{\circ}W$ (not exactly true). Calculate the shortest distance between the two cities, also check the distance if you go from Hong Kong to Havana via the same latitude. (b) Find a city nearest to the antipodal point of Hong Kong.
2. (Not related to great circles) Show that the maximum number of spheres of radius 1 that can be placed touching a fixed sphere of radius 1 so that no pair of spheres has any interior point in common is between 12 and 14.
3. Given three 300° great arcs (parts of three great circles) on a sphere, show that it is impossible that every pairwise great arcs have no common points.
4. Given O and $n(\geq 4)$ points. Suppose for any 3 points, there exist (among the n points) a fourth point so that O is in the interior of the polyhedron (excluding the faces and edges). Determine the possible value(s) of n .
5. Given a surface in the 3-space. Form a circle using any three points of the surface. Suppose the circle lies entirely on the surface, show that the surface is a plane or a sphere.
6. On a spherical planet X there are $2n$ gas stations. Each station is paired with a gas station located on the diametrically opposite point on the planet. Each station has a fixed (non-replenishable) amount of gas. The arrangement of stations and the amount of gas at each station are such that it is possible for a car with an empty (and large enough) reservoir to start from any station and reach its paired station (filling up its reservoir with any gas it encounter along the way). For which n does

it follow that it is possible for the car to start with some gas station and visit all other gas stations on the planet? Assume the car consumes a fixed amount of gas per unit length.

Mini-Research

1. Try one of the problems from Exercise 2 to 6, give some variation if possible, and be honest try a problem you have not seen before.
2. Find a property of great circle and prove it.

Exercises

1. (Warm-Up) Assume Hong Kong's position is $22^\circ N, 114^\circ E$, with the radius of the Earth 6371 km. (a) The position of Havana is $22^\circ N, -82^\circ W$ (not exactly true). Calculate the shortest distance between the two cities, also check the distance if you go from Hong Kong to Havana via the same latitude. (b) Find a city nearest to the antipodal point of Hong Kong.
2. (Not related to great circles) Show that the maximum number of spheres of radius 1 that can be placed touching a fixed sphere of radius 1 so that no pair of spheres has any interior point in common is between 12 and 14.

(Brazilian MO 1983) Place 6 spheres surrounding the fixed sphere, with 3 more at top and 3 at bottom, thus 12 can be achieved. Taking O as the centre of the fixed sphere, another sphere blocks off a solid angle, the angle between the centre line and the surface is 30° . Hence the solid angle block off is $2\pi(1 - \cos \frac{\pi}{6}) = \pi(2 - \sqrt{3})$.

Thus we have at most $\frac{4\pi}{\pi(2 - \sqrt{3})} < 15$ spheres touching the fixed spheres, i.e. at most 14.

3. Given three 300° great arcs (parts of three great circles) on a sphere, show that it is impossible that every pairwise great arcs have no common points.

(First IMO, Beijing, p.196, Math Competition, Sin Jun p.492) By contradiction, ultimately breaks down to the fact that the sum of two plane angles of a triangle is greater than the other angle.)

4. Given O and $n (\geq 4)$ points. Suppose for any 3 points, there exist (among the n points) a fourth point so that O is in the interior of the polyhedron (excluding the faces and edges). Determine the possible value(s) of n .

(Sin Jun p. 494) Use Euler and projection).

5. Given a surface in the 3-space. Form a circle using three points of the surface. Suppose the circle lies entirely on the surface, show that the surface is a plane or a sphere.

(Hard, 463 Russian Problems).

6. On a spherical planet X there are $2n$ gas stations. Each station is paired with a gas station located on the diametrically opposite point on the planet. Each station has a fixed (non-replenishable) amount of gas. The arrangement of stations and the amount of gas at each station are such that it is possible for a car with an empty (and large enough) reservoir to start from any station and reach its paired station (filling up its reservoir with any gas it encounter along the way). For which n does it follow that it is possible for the car to start with some gas station and visit all other gas stations on the planet? Assume the car consumes a fixed amount of gas per unit length.

(Serbian MO2019, $n \leq 3$.)

Russian Serbia Brazil Chinese